

Plasma Cell Dyscrasias

MGUS · Smoldering MM · Multiple Myeloma — IM chalk-talk one-pager (TeachIM, CC BY-NC 4.0; opt. 2026)

One-liner: Clonal plasma-cell proliferation produces a monoclonal immunoglobulin (M-protein/paraprotein). End-organ damage = **CRABI**. Screen with SPEP+immunofixation + serum free light chains (FLC).

CRABI — end-organ damage (defines MM)

Feature	Threshold	Mechanism
Calcium ↑	≥11 mg/dL	RANK-L / osteoclast
Renal	Cr ≥2 mg/dL	Light-chain cast nephropathy
Anemia	Hgb <12 g/dL	Marrow infiltration
Bone	Lytic lesion / fracture	Osteoclast activation
Immunodefic.	Recurrent infection	Hypogammaglobulinemia

Testing — sensitivity ladder

Test	Adds	Sens.
SPEP + IFX	Detect + size + type M-protein	~80%
+ UPEP (24 h)	Bence Jones light chains in urine	~95%
+ serum FLC	Light-chain-only dz; no 24-h collect	~99%

- Dipstick = **albumin only** → misses Bence Jones protein.
- **FLC ratio κ/λ** normal 0.26–1.65; >1.65 κ clone, <0.26 λ clone. Renal failure raises ratio (up to ~3) → use >4 or <0.25.
- ~16% of MM is light-chain-only (SPEP-negative) → always send FLC.

MGUS vs SMM vs MM

	MGUS	SMM	MM
Serum M	<3 g/dL	≥3 g/dL	≥3*
Marrow PC	<10%	10–60%	≥10%
CRABI	None	None	Yes / biomarker
Prog.	~1%/yr	~10%/yr×5y	—

*Oligo-/non-secretory MM may not meet M criterion → dx on marrow/FLC.

- **Myeloma-defining (SLiM)**, any one = MM: marrow ≥60%; FLC ratio ≥100; >1 focal MRI lesion.

Prognosis — R-ISS (send β 2M mg/L, LDH, albumin, FISH)

Stage	Definition (abbrev.)	5-yr OS	Median OS
I	β 2M <3.5 mg/L + nl LDH + std FISH + alb ≥3.5 g/dL	82%	NR
II	Neither I nor III	62%	83 mo
III	β 2M ≥5.5 mg/L AND (↑LDH OR high-risk FISH)	40%	43 mo

High-risk FISH: del(17p), t(4;14), t(14;16). **R2-ISS (2022)** adds gain/amp 1q21.

Treatment & toxicity prophylaxis

Class (ex.)	Signature tox	Prophylaxis
Proteasome inhib. (bortezomib, carfilzomib)	Neuropathy (bort); CV (carfil); cytopenias	VZV acyclovir
IMiD (lenalidomide, pomalidomide)	VTE/arterial clot; cytopenias	ASA or anticoag
Steroid (dex)	Hyperglycemia, infxn, bone loss	PJP ppx, PPI, Ca/vit-D
Anti-CD38 mAb (daratumumab)	Infusion rxn; interferes w/ crossmatch	Premed; type&screen early

- **Induction (2024):** quadruplet = anti-CD38 mAb + VRd (D-VRd) → autoSCT if eligible (age <75, ECOG <3, no cirrhosis/ sev HF; **renal not a contraindication**) → maintenance.

Complication management

- **HyperCa:** IVF + bisphosphonate (± calcitonin).
- **Cast nephropathy:** aggressive IVF to UOP ~3 L/d + correct Ca + **urgent systemic therapy**.
- **Bone:** bisphosphonate + vit-D (Ca if not hyperCa); palliative XRT; avoid surgery unless emergent.
- **Recurrent infxn + IgG <500:** consider IVIG.
- **Anemia:** exclude reversible causes; transfuse as needed.